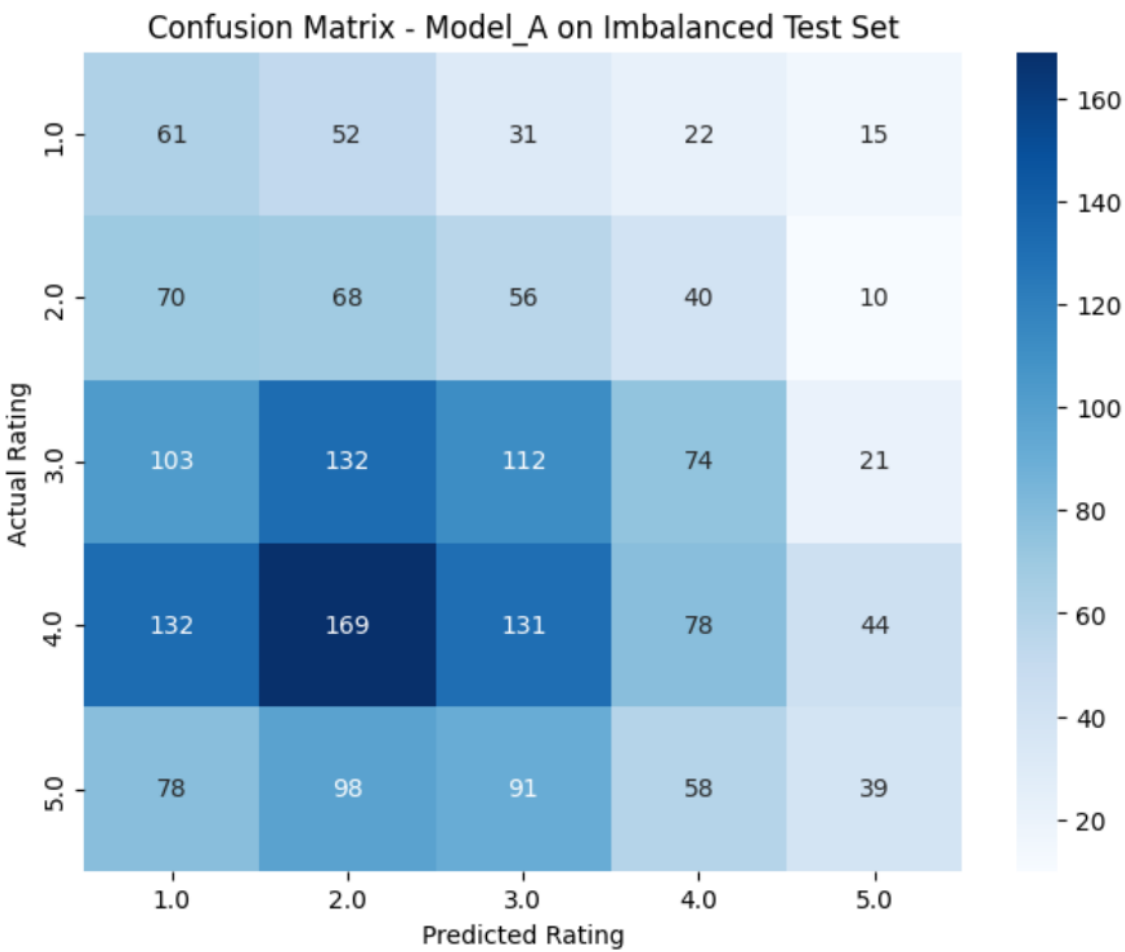


Cross-Testing Report

Summary of Model Performance

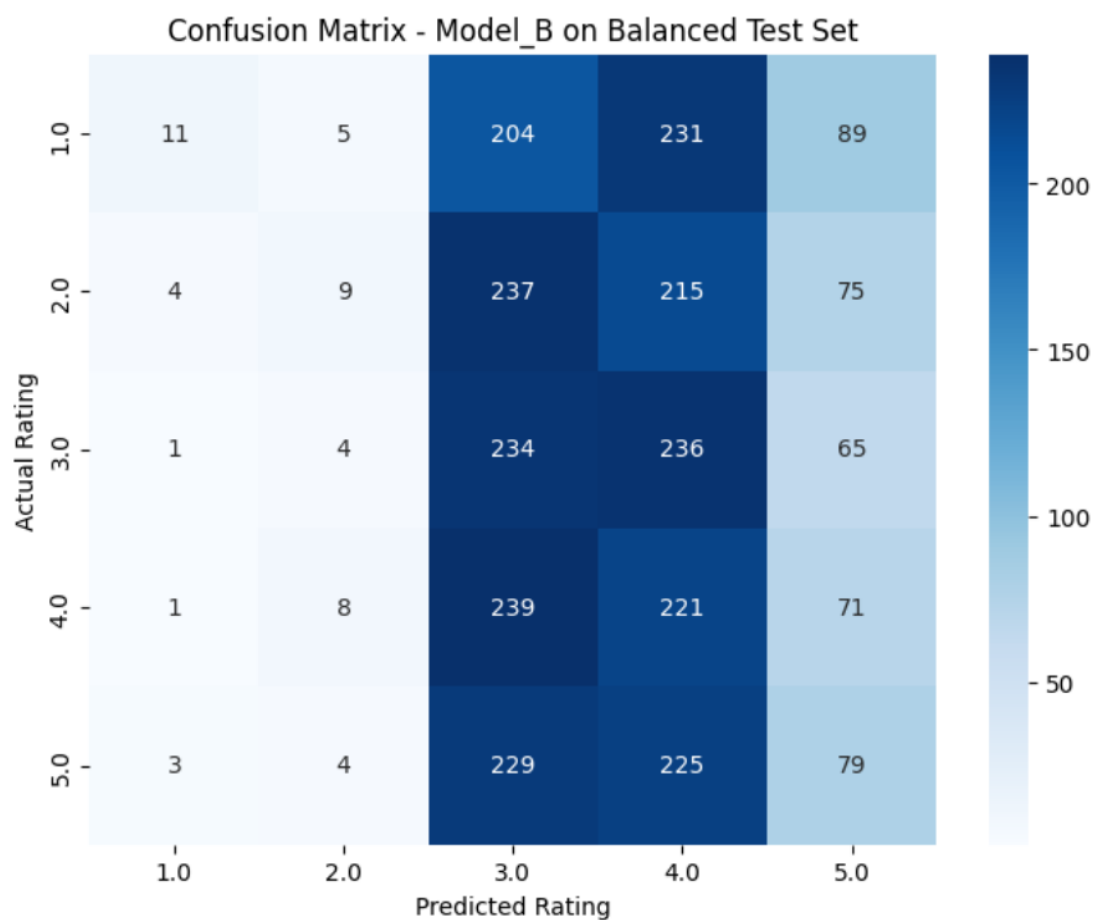
Model_A (Balanced-trained) on Imbalanced Test Set

- Accuracy: 0.2006
- Precision: 0.2484
- Recall: 0.2006
- F1-Score: 0.1993
- The model shows poor generalization when tested on an imbalanced dataset, with all metrics close to random guessing. Performance varies by class, but most ratings are predicted incorrectly, as seen in the confusion matrix.



Model_B (Imbalanced-trained) on Balanced Test Set

- Accuracy: 0.2052
- Precision: 0.2918
- Recall: 0.2052
- F1-Score: 0.1572
- Similarly, the imbalanced-trained model performs poorly on a balanced dataset, showing a heavy bias toward predicting certain classes and failing to capture the true distribution of balanced ratings.



Observations on Data Distribution and Generalization

1. Training Data Distribution Matters – Both models struggle when tested on a dataset with a different distribution than they were trained on.
2. Balanced Training Improves Minority Class Recognition – Model_A trained on balanced data predicts minority classes better in balanced scenarios but fails when faced with skewed distributions.

3. Imbalanced Training Causes Bias – Model_B tends to overpredict majority classes, leading to poor results in balanced datasets.
4. Cross-Testing Shows Domain Shift Problem – Both models face a sharp performance drop when the training and testing data distributions do not match.

Based on the results:

- Neither model performs well under cross-distribution testing, but Model_A (Balanced-trained) has slightly better balanced-class handling and could be adapted with reweighting or fine-tuning for imbalanced scenarios.
- If deployment data is expected to be imbalanced, retraining with imbalanced-aware techniques (e.g., class weighting, SMOTE oversampling) is recommended.
- If deployment data is expected to be balanced, use Model_A directly but monitor performance on real-world data.

Final Choice: Model_A with further improvements, since it generalizes slightly better across classes and is less biased toward majority ratings compared to Model_B.